

Zachary A. Sexton, *PhD*

Bioengineering

Computational Modeling & Simulation
Scientist · Engineer

CONTACT

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EDUCATION

PhD, Bioengineering
Stanford University
2024 · Advisors: Alison Marsden, Sean Wu, Mark Skylar-Scott.

MS, Bioengineering
Stanford University
2021

BE, Biomedical Engineering · BA, Public Policy Analysis
University of Delaware
2018

TECHNICAL SKILLS

Modeling & numerical methods
CFD / FSI / structural mechanics · finite-element simulation · tetrahedral meshing · Navier–Stokes hemodynamics · Darcy perfusion · OD/1D/3D reduced- & full-order modeling · lumped-parameter networks · surrogate-based optimization · parameter estimation · boundary-condition calibration · mesh / time-step convergence

Programming & software
Python · Cython · C++ · Fusion 360 · COMSOL · svMultiPhysics (in-house multiphysics solver) · Linux · Windows · HPC (SLURM) & distributed compute

Geometry, fabrication & data
STL & mesh model preparation · synthetic vascular tree generation for whole-organ modeling · watertight networks for CFD and 3D fabrication · CFD post-processing · image-analysis scripts · open data/model release · bioprinting workflow integration

Experimental & validation
3D extrusion bioprinting · tissue-engineered perfusion experiments · cell-viability analysis · clinical CT myocardial perfusion imaging · patient-specific cardiovascular modeling

PROFILE

Recent Stanford Bioengineering PhD and postdoctoral scholar with 6+ years of post-baccalaureate research experience building computational models and multiphysics simulation software for complex fluid, transport, vascular, and biomanufacturing systems. Specializes in computational fluid dynamics, finite-element hemodynamics, multiscale coronary/myocardial blood-flow models, synthetic vascular generation, and model-guided design workflows. Strong fit for roles requiring mesh-based simulation, solver workflow automation, model validation and calibration, design exploration, optimization, and clear technical communication.

CAPABILITIES

Multiscale & multiphysics modeling	Built multi-scale, multi-fidelity OD/3D hemodynamic models, Darcy myocardial perfusion models, lumped-parameter cardiovascular networks, synthetic vascular trees, and coupled CFD/perfusion workflows.
Finite-element simulation	Developed and applied multiphysics FEM with tetrahedral meshes; incorporated mesh and time-step convergence checks for complex clinical geometries of aneurysm CFD cases.
Python simulation software	Author of svVascularize (svv), an open-source Python API for automated vascular generation and multi-fidelity hemodynamic simulation for large-scale tissue and organ biomanufacturing.
Design exploration & optimization	Built model-guided vascular design workflows, surrogate-based parameter estimation, and iterative tuning to satisfy pressure, flow, perfusion, and geometry constraints for biologic and soft-matter applications.
Validation & calibration	Calibrated and evaluated models against clinical CT perfusion / cardiac-function measurements and bioprinted tissue perfusion / cell-viability experiments.
Communication & collaboration	Lead author on a Science paper; collaborated across computational biomechanics, cardiology, bioengineering, and 3D bioprinting teams; released software and data for reproducibility.

APPOINTMENTS

Postdoctoral Scholar, Cardiology Stanford University · Stanford, CA	2024 – Present
Cardiovascular physiology, disease modeling, and tissue engineering through stem-cell biology, hemodynamics, 3D extrusion bioprinting, and CFD. Builds model-guided workflows linking synthetic vascular design, fluid simulation, and bioprinting to improve cardiac tissue-engineering strategies for disease modeling and therapeutic development. Affiliated with the Cardiovascular Biomechanics Computation Lab and Wu Lab; collaborates across computational, biological, and fabrication teams.	
Doctoral Researcher, Bioengineering PhD Stanford University · Stanford, CA	2018 – 2024
Dissertation: Synthetic Vascular Design and Simulations for Biomanufactured Tissues and Patient-Specific Modeling Applications. Developed algorithms and workflows for rapid synthetic vascular network generation in arbitrary geometries, multi-fidelity hemodynamic simulations, computational vasculature modeling, and 3D fabrication preparation. Defined model objectives, boundary conditions, operating assumptions, and performance targets for pressure, flow, perfusion, wall shear stress, and tissue-viability outcomes. Used computational modeling alongside experiments and clinical imaging to calibrate, validate, and interpret model behavior.	

HONORS & AWARDS

NSF Graduate Research Fellowship
2018

EDGE-STEM Fellow, Stanford VPGE
2018

Bioengineering TA Award, Stanford School of
Engineering
2019–20

Alexander J. Taylor Sr. Award, University of
Delaware
2018

Goldwater Scholarship — Honorable Mention
2017

SOFTWARE & OPEN DATA

svVascularize
*Open-source software for vascular generation
and multi-fidelity hemodynamic simulation*
Author/developer; published through the
SimVascular ecosystem and PyPI. Supports
automated de-novo vascular generation for
tissue-engineering and CFD analyses.
2025 — Present

**Supplemental Data — Organ-Scale
Synthetic Vasculature**
*Open Zenodo dataset accompanying the
Science article*

Geometries, CFD results, image-analysis
workflows, and post-processing scripts to
reproduce figures and support new design
studies.
2025

DOCTORAL ADVISORS

Alison Marsden · Sean Wu · Mark Skylar-Scott

SELECTED PROJECTS

Model-Guided Organ-Scale Synthetic Vasculature for Biomanufacturing 2023 — 2025

Lead author on a model-driven design platform combining accelerated synthetic vascular tree generation, multi-fidelity CFD/hemodynamic simulation, and 3D bioprinting.

Demonstrated model-guided vascular perfusion and experimental cell-viability analysis in fabricated living-tissue constructs.

Authored svVascularize (svv), a Python open-source package/API for automated vascular generation and multi-fidelity hemodynamic simulation for tissue & organ engineering.

Personalized Coronary & Myocardial Blood-Flow Modeling 2023 — 2024

Contributed modeling tools to a multi-scale patient-specific flow-simulation framework combining image-based coronary anatomies, synthetic vasculature for sub-resolution arteries, FEM CFD, Darcy myocardial perfusion, and lumped-parameter systemic circulation.

Used clinical CT perfusion imaging and cardiac-function measurements to personalize flow distributions and model parameters via surrogate-based optimization; compared boundary-condition strategies and synthetic-tree variants.

Stack: 3D finite-element Navier–Stokes, TetGen tetrahedral meshes, SimVascular/svSolver, svZeroDPlus coupling, and finite-element Darcy perfusion in svMultiPhysics.

Hemodynamic Predictors of Cerebral Aneurysm Growth 2023

Contributed to matched-pair CFD study of growing vs. stable cerebral aneurysms from MRA/CTA images; evaluated wall shear stress, oscillatory shear index, low-shear area, and mesh-resolution effects.

Emphasized robust modeling, temporal/spatial convergence checks, statistical comparison of hemodynamic predictors, and clear interpretation of simulation uncertainty.

Medical Technology Policy Analytics 2023

Co-authored JAMA Health Forum analysis of FDA authorization-to-Medicare coverage timing for novel technologies, translating quantitative findings for policy and industry stakeholders.

SELECTED PUBLICATIONS

[1] **Sexton, Z. A.**, Rüttsche, D., Herrmann, J. E., Hudson, A. R., Sinha, S., Du, J., Shiwerski, D. J., Masaltseva, A., Solberg, F. S., Pham, J., Szafron, J. M., Wu, S. M., Feinberg, A. W., Skylar-Scott, M. A., & Marsden, A. L. (2025). *Rapid model-guided design of organ-scale synthetic vasculature for biomanufacturing*. **Science**, 388(6752), 1198–1204. doi:10.1126/science.adj6152

[2] Menon, K., Khan, M. O., **Sexton, Z. A.**, Richter, J., Nguyen, P. K., Malik, S. B., Boyd, J., Nieman, K., & Marsden, A. L. (2024). *Personalized coronary and myocardial blood-flow models incorporating CT perfusion imaging and synthetic vascular trees*. **npj Imaging**, 2(1), 9. doi:10.1038/s44303-024-00014-6

[3] Brown, A. L., **Sexton, Z. A.**, Hu, Z., Yang, W., & Marsden, A. L. (2024). *Computational approaches for mechanobiology in cardiovascular development and diseases*. **Current Topics in Developmental Biology**, 156, 19–50. doi:10.1016/bs.ctdb.2024.01.006

[4] Weiss, A. J., Panduro, A. O., Schwarz, E. L., **Sexton, Z. A.**, Lan, I. S., Geisbush, T. R., Marsden, A. L., & Telischak, N. A. (2023). *A matched-pair case-control study identifying hemodynamic predictors of cerebral aneurysm growth using CFD*. **Frontiers in Physiology**, 14, 1300754. doi:10.3389/fphys.2023.1300754

[5] **Sexton, Z. A.**, Perl, J. R., Saul, H. R., Trotsyuk, A. A., Pietzsch, J. B., Ruggles, S. W., Nikolov, M. C., Schulman, K. A., & Makower, J. (2023). *Time from FDA authorization to Medicare coverage for novel technologies*. **JAMA Health Forum**, 4(8), e232260. doi:10.1001/jamahealthforum.2023.2260

[6] Puluca, N., Münsterer, A., Prinzing, A., **Sexton, Z. A.**, Lange, R., Meyer-Saraei, R., & Scharfshwerdt, M. (2020). *In-vitro comparison of everting vs. non-everting suture techniques for a supra-annular biological heart valve*. **Journal of Thoracic Disease**, 12(5), 2443–2449. doi:10.21037/jtd.2020.03.55